

Competitive analysis & differentiation

*Purpose: show that this project was designed after researching what already exists, that we understand the gaps, and that our differentiation is a **specific, defensible combination** — not the false claim "nobody has built a dementia app".*

Last researched: 2026-09-09.

1. What already exists

Face recognition for family members — EXISTS

Product	What it does	Notes
Timeless (Timeless Innovations)	"Identify" feature: point camera → name + relationship. Also photo feed, day reminders, picture phone book.	Built by a teenager for a grandparent; consumer app. Faces auto-tagged by AI.
I-Remember (Microsoft / Univ. of Lisbon)	AI face recognition to help early-stage patients recognise people.	Research/pilot.
"Remember Me!"	Uploaded photos + GPS: alerts patient when a known contact approaches, shows who + photo.	Concept/prototype.
Various academic apps	Real-time face recognition, add faces live or by image.	Papers, not shipped products.

Takeaway: face-ID for family is **not novel on its own**. We must not claim it is.

GPS tracking & geofencing — EXISTS, mature, mostly hardware

AngelSense, SureSafe, Family1st, MedicAlert — dedicated GPS watches/pendants with live location, safe-zone geofencing, exit alerts, SOS button. This is a solved, commoditised space and the incumbents have better hardware battery life than a phone app ever will.

Takeaway: our geofencing is table-stakes, not a differentiator. Be honest that dedicated trackers are better for pure safety; our value is *integration* with the rest of the context.

Medication reminders — COMMODITY

Medisafe, MyTherapy, plus every dementia app bundles reminders. Nothing to claim here.

Caregiver coordination — EXISTS

Carely (private family feed, visit coordination, shared updates), CareMobi (connects caregivers + day centres + providers), Alzheimer's Association "My ALZ Journey" (2025, newly-diagnosed navigation). These are communication/logistics tools, not AI memory systems.

AI chatbots in dementia care — EXISTS, but aimed at the CAREGIVER, grounded in CLINICAL text

System	Who it helps	Knowledge source
RISE (social robot + RAG)	Caregiver — training, stress management	Verified <i>REACH Caregiver Notebook</i> . Reports 87% correctness, 92% relevancy. Explicitly uses RAG to minimise hallucination.
ADQueryAid	ADRD caregivers — Q&A, emotional support	LLM + RAG over ADRD knowledge; beat ChatGPT-3.5 on usability.

System	Who it helps	Knowledge source
AlzheimerRAG	Clinicians / researchers	Multimodal RAG over PubMed articles.
DEMENTIA-PLAN	Care planning	Multi-knowledge-graph RAG.

Takeaway: "RAG to reduce hallucination in dementia AI" is a **known research idea** — but every instance grounds on a *fixed, generic, clinical* corpus and serves the *caregiver*.

Integrated "all-in-one" platforms — EMERGING in research, still fragmented in market

CareBuddy / "mobile care ecosystem" (assessments + AI chatbot + GPS + telemedicine + peer support) and Demensia KITA ("one-stop centre") are academic prototypes. Systematic reviews say commercial apps remain **fragmented**, lack **personalization**, and struggle with the tension between **simple** and **multifunctional**.

2. Gaps we found (and target)

#	Gap in the market	Our answer
G1	AI dementia assistants retrieve from generic clinical text to help the caregiver . No one runs RAG over the family's own curated memories to answer the patient's "tell me about Rahul".	Personal-memory RAG for the patient. The knowledge base is per-patient, human-written family memories — not PubMed.
G2	Face-ID apps "learn faces" and photo-feed apps let family post freely — but the AI is not gated on human-verified facts . Research RAG systems ground on a <i>fixed</i> notebook nobody edits.	Caregiver-controlled knowledge with an approval queue. <code>memories.status</code> = pending/approved/rejected. The assistant can only say approved facts. AI-suggested memories wait for caregiver approval.
G3	Consumer dementia apps give answers with no provenance . Explainability is a research-only concept.	Source attribution shown in-app: "Based on: a memory from 12 March — 'Rahul visited with mangoes'."
G4	A confidently wrong answer to a dementia patient is uniquely harmful, yet consumer assistants happily guess.	Hallucination-resistant by design: similarity floor, approved-only retrieval, explicit "I'm not sure — ask a family member" fallback. Never invents relationships or events.
G5	Capabilities are split across apps and devices : face-ID in one app, GPS in a watch, reminders in another, coordination in a fourth. Patient must switch contexts; caregiver juggles dashboards.	One Context Engine fuses face + relationship graph + approved memories + medication + location into a <i>single</i> calm response. "Who is this?" → who + how they're related + when they last visited. "Why am I here?" → time + place + next medication + today's routine.
G6	Biometric data of vulnerable people is captured casually ("the app learns faces").	Consent-first: explicit consent record before any face is registered; we store the embedding, not the photo ; one-tap delete of a person removes their embeddings + memories.
G7	Personal relationships aren't modelled — apps store a flat contact list.	Relationship graph (self-referential table): "Rahul → son of → patient", "Meera → married to → Rahul" feeds richer context.

3. Honest positioning (use this in interviews)

*"I researched the space first. Face recognition for family exists — Timeless, Microsoft's I-Remember. GPS geofencing is a mature hardware market — AngelSense, SureSafe. AI chatbots for dementia exist in research — RISE, ADQueryAid — but they help the caregiver and are grounded in medical literature. Two gaps stood out. **One**, the market is fragmented — each capability is a separate app or device. **Two**, where AI exists, no human controls what it tells the patient. So my project's thesis is a single system where the assistant's entire knowledge is a **caregiver-approved, per-patient memory store**; every answer is **grounded and cites its source**; it **refuses to guess**; and a **Context Engine** fuses recognition, relationships, memories, medication and location into one response. I'm not claiming I invented the dementia app. I'm claiming this specific combination — caregiver-controlled, hallucination-resistant, context-aware personal memory — is not something the current market or research prototypes actually ship."*

4. What NOT to claim (interview traps)

- [x] "Nothing like this exists." → False; you'll be caught. Say "this *combination* isn't shipped."
- [x] "Better than AngelSense for safety." → Dedicated trackers have better battery/hardware. Our edge is integration, not raw tracking.
- [x] "Medically validated." → It's a prototype, not a medical device. Always say so.
- [x] "Face recognition is my innovation." → It isn't. The *caregiver-controlled, consent-first, context-fused* use of it is the story.

5. Sources

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